

Task	Approach	Examples	Epochs	2×V100	1×L4
Granulometry	Direct	18	40	24m	19m
Granulometry	CoT-aug	72	40	101m	77m
Steel	Direct	30	40	40m	20m
Steel	CoT-aug	120	40	170m	85m
UHCS	Direct	30	40	58m	27m
UHCS	CoT-aug	120	40	240m	111m
Weld	Direct	24	40	30m	15m
Weld	CoT-aug	96	40	125m	62m

Table 3: Training time per task and approach.

training times, we ran 2 epochs of each configuration on a single NVIDIA L4 24GB GPU (GCP Vertex AI) and extrapolated to 40 epochs. The L4 times are faster due to avoiding inter-GPU communication overhead.

4.4 Evaluation Protocol

Evaluation uses the held-out test split, never seen during training or CoT generation. Each test image is processed with the full classification prompt at temperature 0.1. We set max new tokens to 256. The JSON is parsed from the model output; if the model produces a description followed by JSON, the JSON is extracted from within the text.

For granulometry, we report accuracy on both axes separately (size, grading) and their combination (both correct). For the other three tasks, we report overall classification accuracy. All metrics are computed on 100% of the test set.

5 Results

5.1 Cross-Task Comparison

Table 4 presents the main results across all four tasks. Our answer-conditioned CoT distillation (CoT-Aug) outperforms Direct LoRA on every task, with improvements ranging from 0.9 to 8.3 percentage points. On three of four tasks, the fine-tuned 3B model also outperforms the frontier model (GPT-4.1) in its few-shot setting.

Task	Cls	Train	Base ZS	GPT-4.1 FS	Direct	CoT-Aug
Granulometry	9	18	12.0%	29.6%	71.3%	79.6%
Steel Surface	6	30	21.7%	91.1%	63.1%	66.7%
UHCS	5	30	60.8%	71.7%	65.8%	66.7%
Weld Defects	4	24	30.8%	65.0%	73.3%	75.8%

Table 4: Main results across four tasks. Granulometry reports “both correct.” Bold = best.

- Three observations stand out:
- 1. CoT distillation consistently helps.** The augmented approach beats Direct LoRA on every task, ranging from +0.9pp (UHCS) to +8.3pp (granulometry). The largest gains appear where the task has more classes and more fine-grained visual distinctions.
 - 2. Small models beat the frontier model.** On weld defects, our 3B model outperforms GPT-4.1 few-shot by 10.8 percentage points (75.8% vs 65.0%). On granulometry, the gap